

MAHESH SARAN

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EDUCATION

Jodhpur Institute of Engineering and Technology
B.Tech. in CSE & AIML

2024 – 2028
CGPA: 7.9

R.K. Paradise ACA Sr. Sec. School, Pipar City, Jodhpur
Senior Secondary Education

2024
Percentage: 85%

SKILLS

- **Professional Skills:** Project Management, Communication, Self-Discipline, Agile
- **Software/Tools:** Git, GitHub, Linux, Windows, macOS, Visual Studio, Android Studio, VS Code, .NET, Microsoft Office
- **Programming Languages:** Java, Python, C, C++, C#, SQL, Bash Scripting, Shell Scripting, HTML, CSS, JavaScript
- **Programming Concepts:** Object-Oriented Programming, Data Structures & Algorithms
- **AI/ML:** Machine Learning, Deep Learning, Natural Language Processing

TECHNICAL EXPERIENCE

Anand Rathi IT Pvt. Ltd. — .NET Framework Intern

May 2026 – Jun 2026

- Developed a software license management system using C# and .NET Framework to generate and validate license keys for controlled application access.
- Implemented device-specific license validation by binding activated license keys to the system's NIC MAC address, restricting subsequent usage to the registered device.
- Integrated MySQL database operations for storing and validating license information, and implemented application components using the .NET MVC architecture.

Key Technologies: C#, .NET Framework, MVC, MySQL

Jodhpur Institute of Engineering and Technology — Python Developer Intern

In-House Summer Training

Jun – Jul 2025

- Developed a maze game using Python and Pygame, implementing event-driven programming, graphics rendering, and interactive game mechanics.
- Implemented ball movement and collision detection, designing core game logic and mechanics for multiple game levels.
- Applied asynchronous programming concepts using Python's asyncio module and collaborated with team members to debug code and implement game functionality.

Key Technologies: Python, Pygame, asyncio

PROJECTS EXPERIENCE

Real-Time Barge-In Detection for Voice AI — *Personal Project*

- Developed a real-time voice AI pipeline integrating Voice Activity Detection, Speech-to-Text, LLM response generation, and Text-to-Speech into a continuous conversational loop, with real-time audio monitoring for interruption handling.
- Implemented a **barge-in detection mechanism** using **Silero VAD** that continuously monitors microphone input while AI-generated speech is playing, immediately stopping TTS playback when user speech is detected.
- Designed a streaming response pipeline using **Groq Whisper** for speech recognition and LLM token streaming, processing generated responses sentence-by-sentence for TTS playback instead of waiting for the complete response.

Key Technologies: Python, Silero VAD, Groq Whisper, LLM APIs, Edge TTS, Pygame, SoundDevice

License Key Management System — **Anand Rathi IT Pvt. Ltd.** — *Internship Project*

- Developed a software license management system using **C# and .NET Framework** to generate and validate license keys, controlling application access based on valid license credentials and registered system information.
- Implemented device-specific license validation by associating an activated license with the system's **NIC MAC address** during first-time activation, restricting subsequent license usage to the registered device.
- Integrated **MySQL** database functionality for storing and validating license information and implemented the application using the .NET MVC architecture.

Key Technologies: C#, .NET Framework, MVC, MySQL

GUI Resume Builder — *Academic Project*

- Developed a desktop-based resume builder using **Python and Tkinter**, allowing users to enter and manage personal, educational, professional, and project information through a graphical interface and generate formatted resumes.
- Implemented automated **Word and PDF document generation** using **python-docx** and **ReportLab**, with JSON-based data handling and file-management functionality for processing user information.
- Integrated **Pillow** for image handling and designed the application workflow to convert user-provided information into structured Word and PDF resumes, reducing manual document formatting.

Key Technologies: Python, Tkinter, python-docx, ReportLab, Pillow, JSON